

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

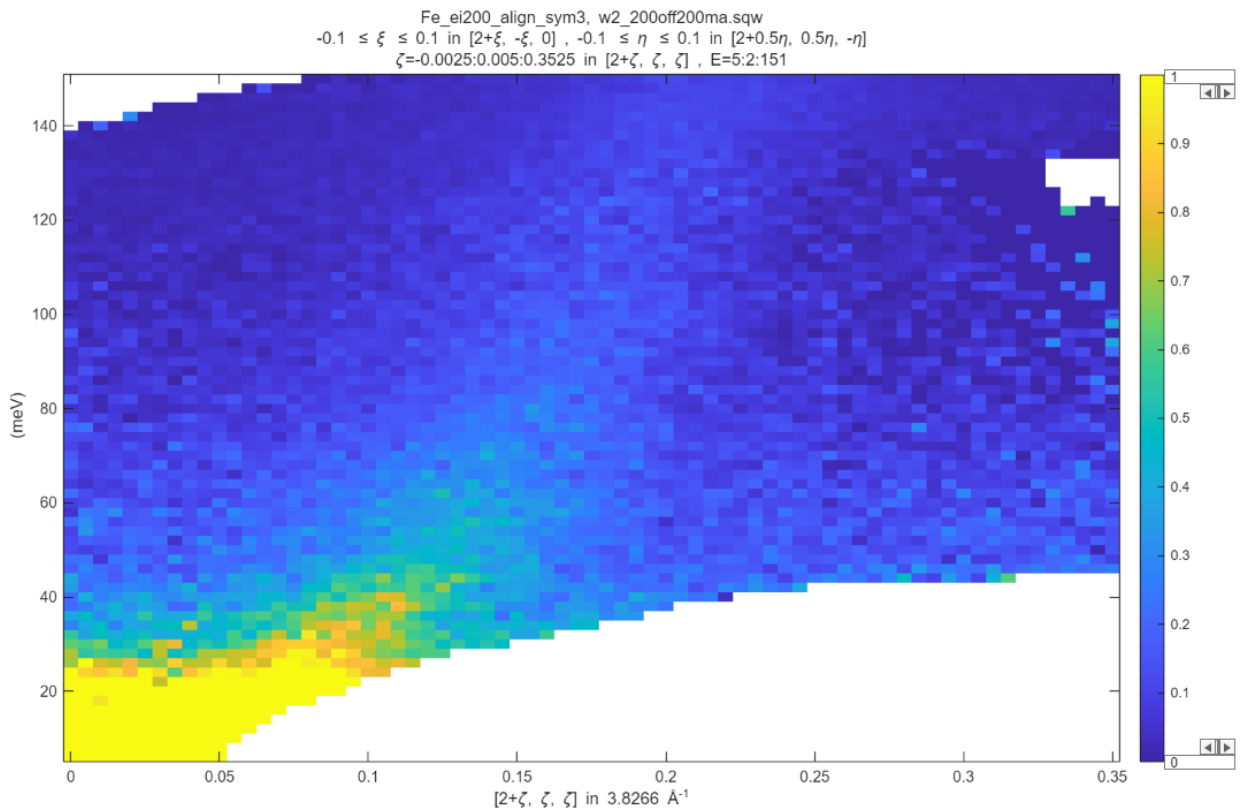
```

```

selectCuts= 'Ei200_off200GPmaff';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end

mi = maps_instrument(200,600,'S');
%mi = maps_instrument(401,600,'S');
%mi = maps_instrument(786,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 110;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

```

```

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

```

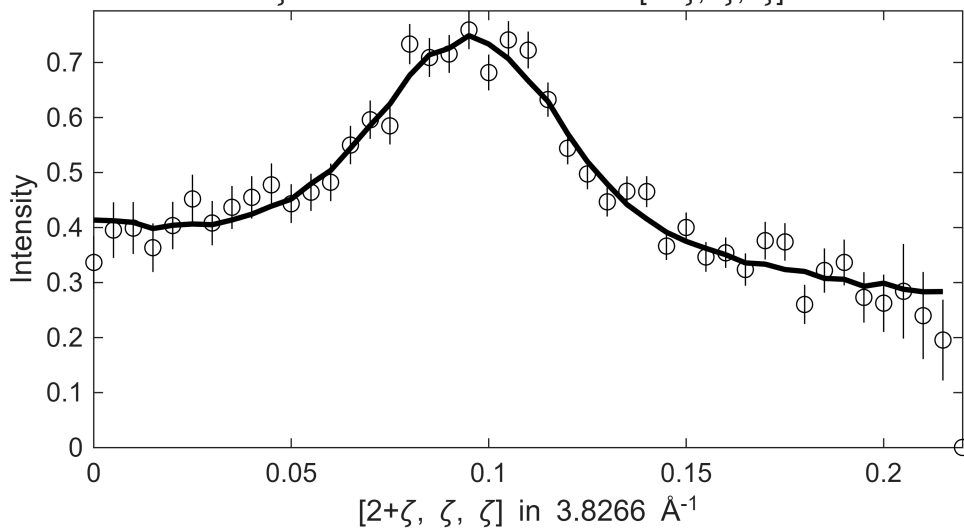
**** En = 35±5 going to 145

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.4085

Fe_ei200_align_sym3, w2_200off200ma.sqw

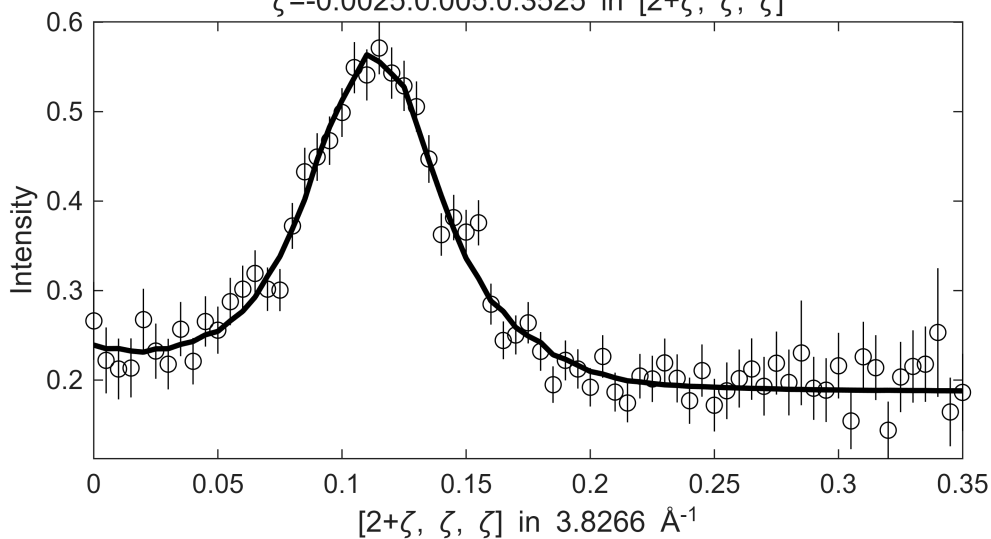
S=0.062±0.0069; J0= 7.4±0.059; gamma=3.09549±1.02212

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $30 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



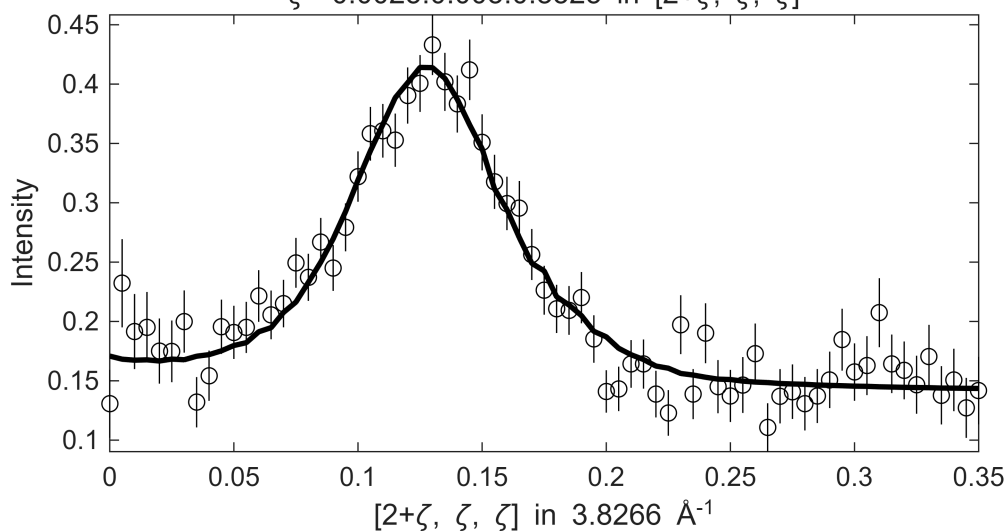
**** En = 45±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.047\pm0.0011$; $J_0= 6.4\pm0.035$; $\gamma=3.13529\pm0.0544458$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $40 \leq E \leq 40$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



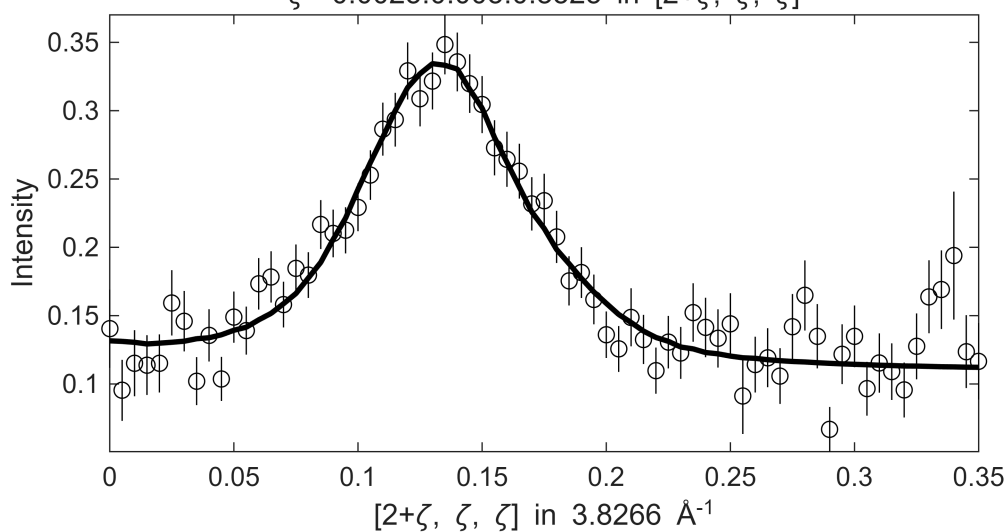
 **** En = 55±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.04\pm0.0012$; $J_0= 5.8\pm0.044$; $\gamma=5.60536\pm0.12448$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $50 \leq E \leq 50$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



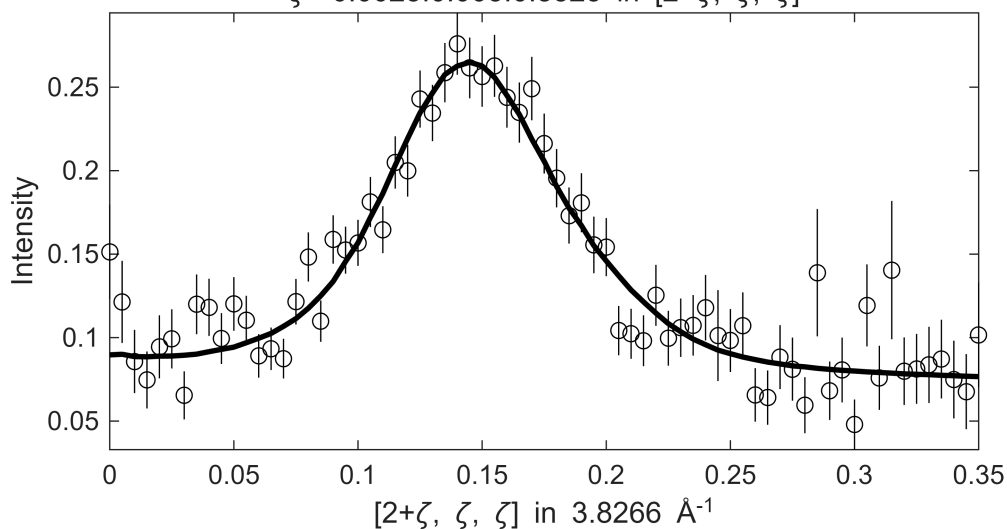
 **** En = 65±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.035\pm0.0025$; $J_0= 5.6\pm0.044$; $\gamma=7.61591\pm1.32187$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq 60$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



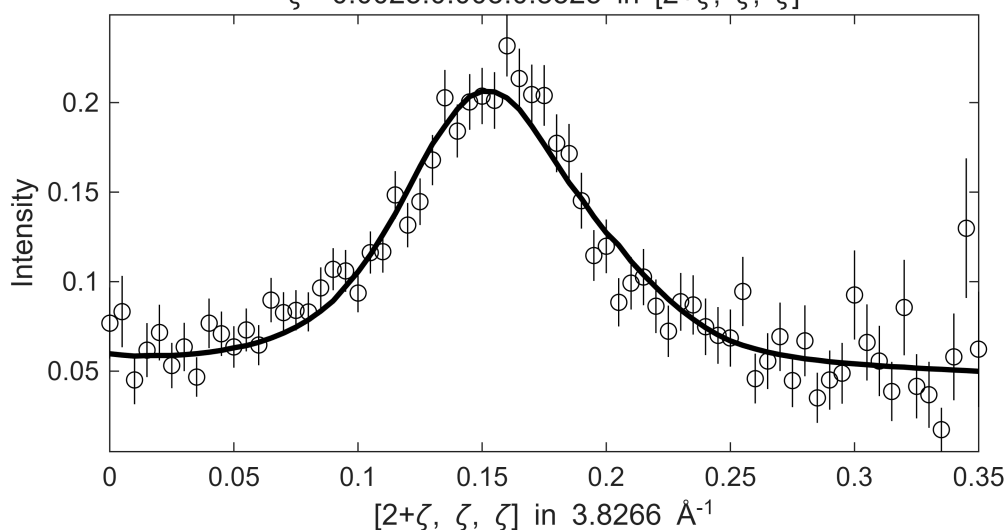
 **** En = 75±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.033\pm0.0027$; $J_0= 5.2\pm0.043$; $\gamma=10.1855\pm1.72047$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $70 \leq E \leq 70$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



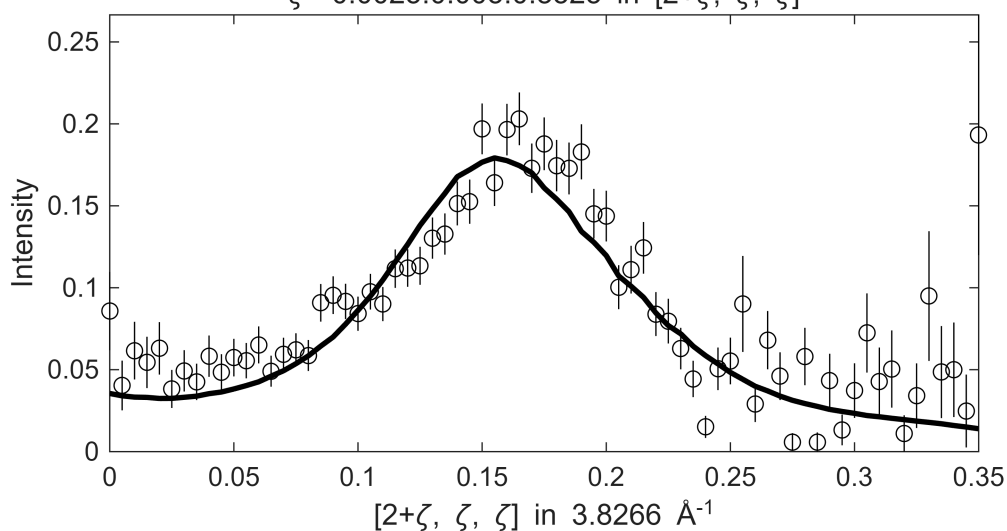
 **** En = 85±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.027\pm0.0012$; $J_0=5\pm0.042$; $\gamma=12.6737\pm0.485717$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $80 \leq E \leq 80$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



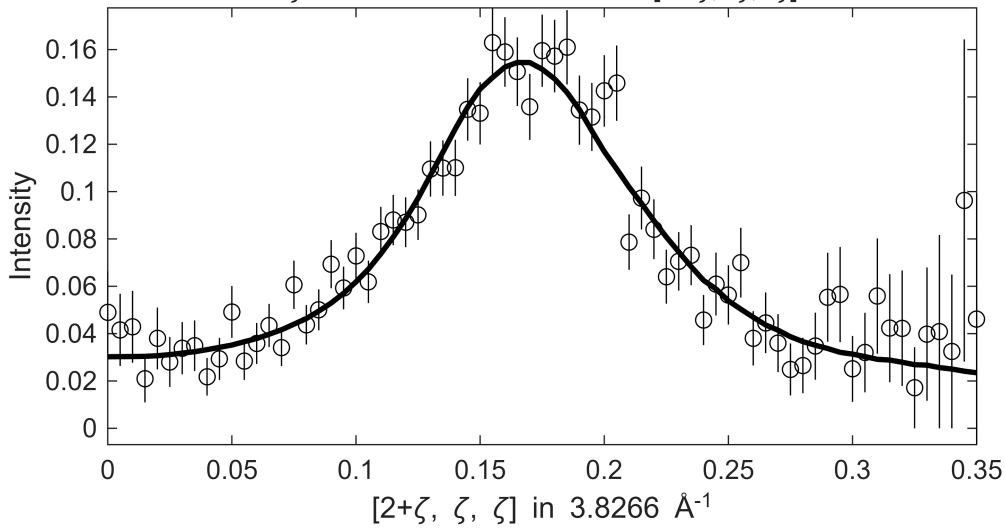
 **** En = 95±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.033\pm0.0044$; $J_0=4.9\pm0.073$; $\gamma=21.5913\pm4.52606$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $90 \leq E \leq 90$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



 **** En = 105±5 going to 145

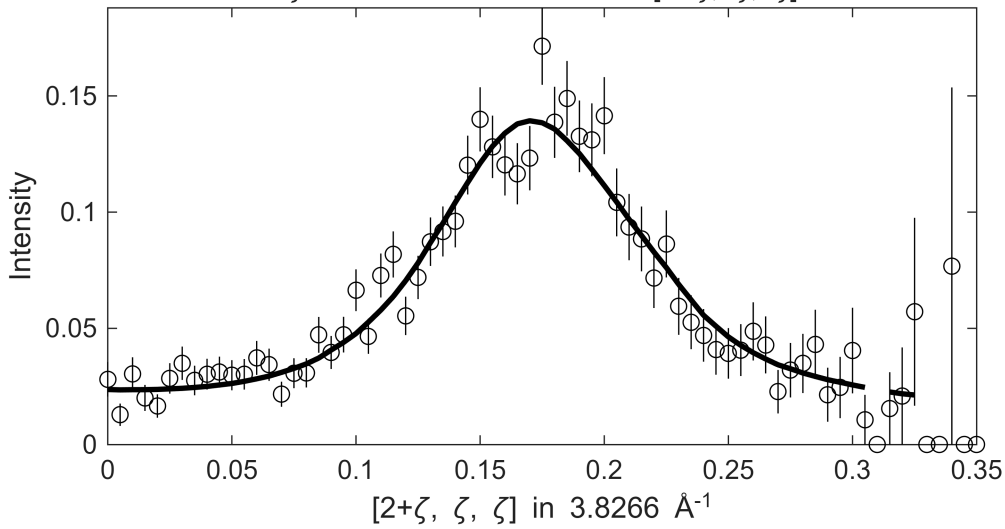
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.022\pm0.0016$; $J0= 4.5\pm0.035$; $\gamma=18.3198\pm1.8646$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $100 \leq E \leq 110$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



 **** En = 115±5 going to 145

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.0423

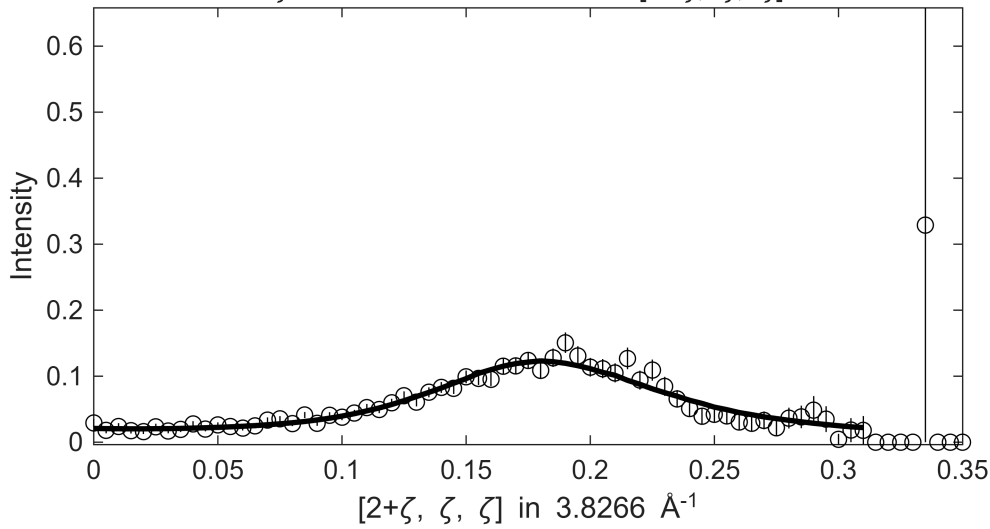
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.017\pm0.00095$; $J0= 4.5\pm0.042$; $\gamma=19.5791\pm1.54659$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $110 \leq E \leq 125$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



 **** En = 125±5 going to 145

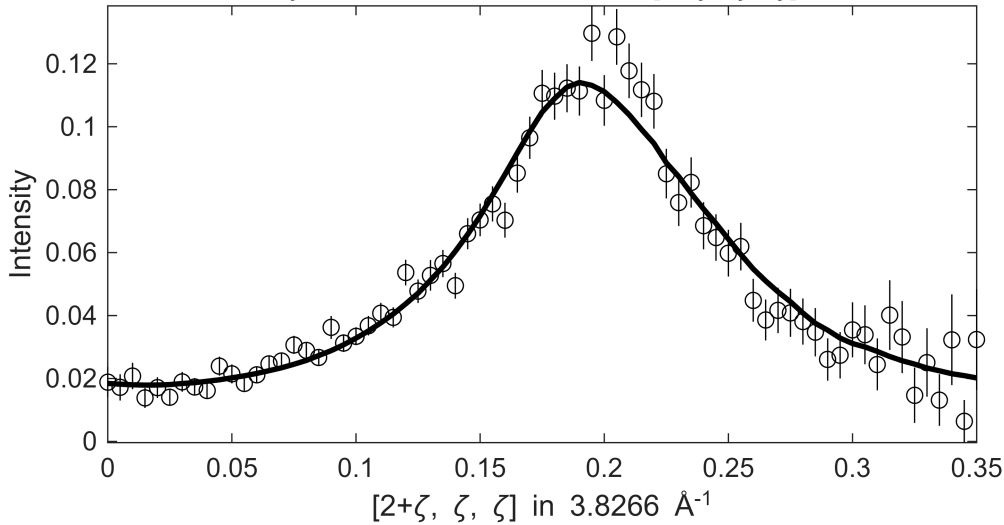
Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 9.8592

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.02\pm0.0013$; $J_0= 4.5\pm0.047$; $\gamma=27.2806\pm3.08845$
 $1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $120 \leq E \leq 130$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



 **** En = 135±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.027\pm0.0012$; $J_0= 4.1\pm0.024$; $\gamma=25.8027\pm1.92488$
 $1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $130 \leq E \leq 140$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



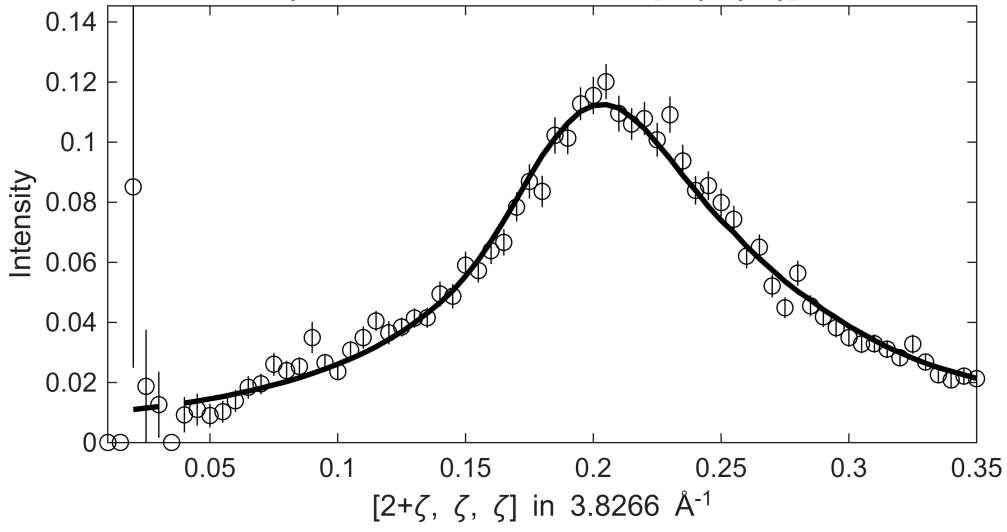
 **** En = 145±5 going to 145

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.2254

Fe_ei200_align_sym3, w2_200off200ma.sqw

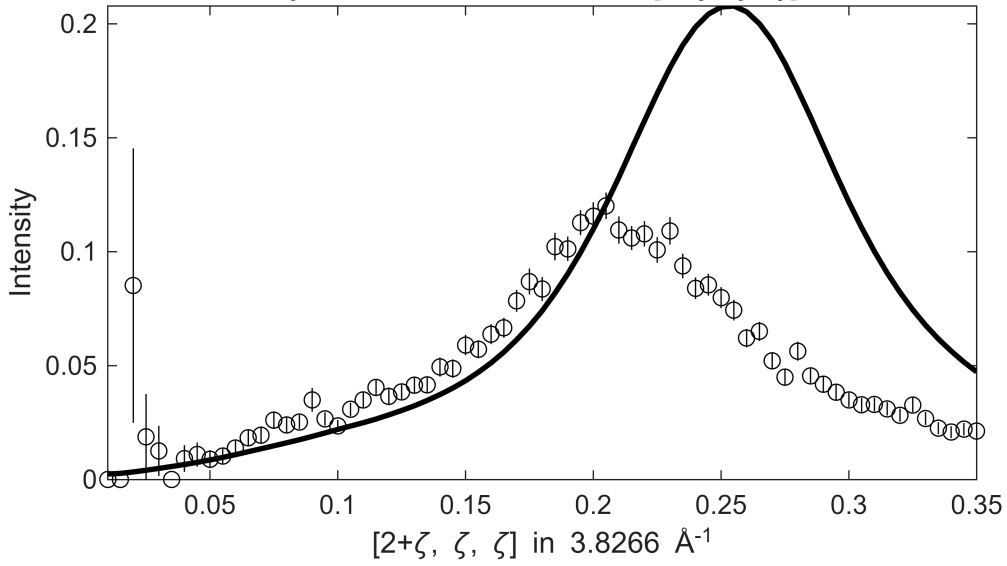
$S=0.03\pm0.0014$; $J_0= 3.8\pm0.016$; $\gamma=27.8313\pm1.97196$

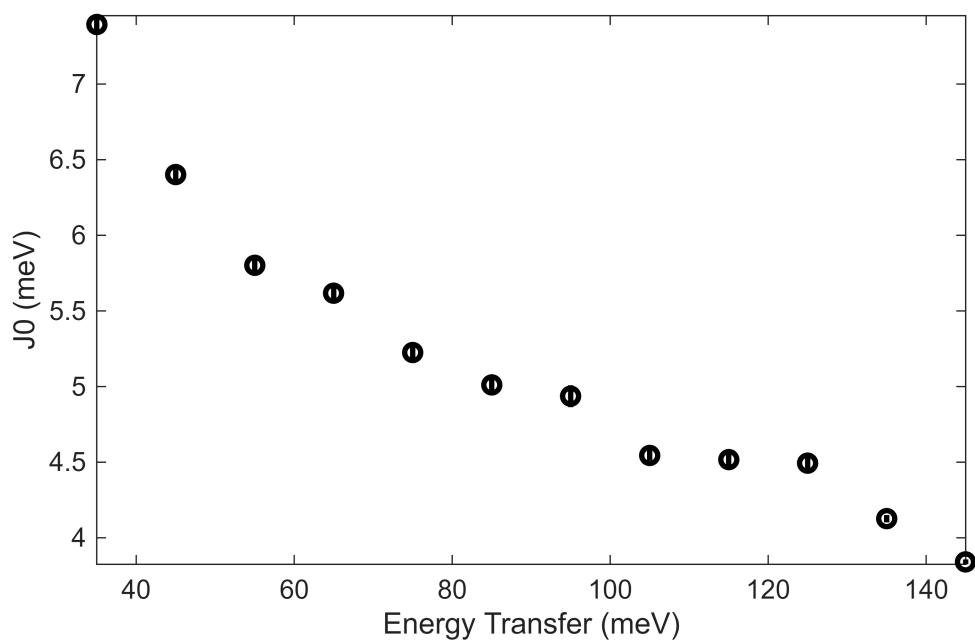
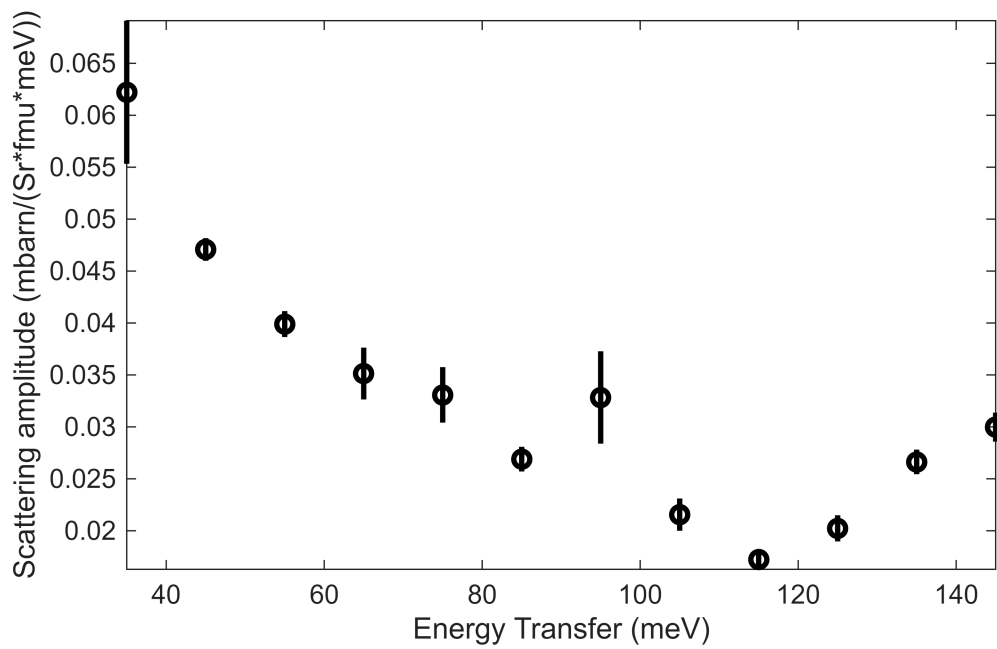
$1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $140 \leq E \leq 150$ eV
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

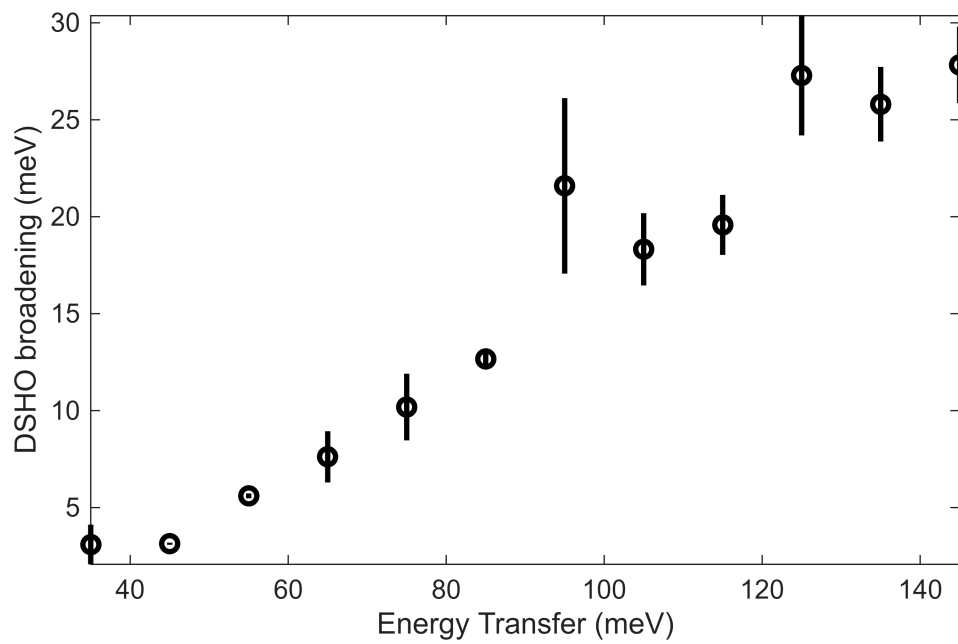


Fe_ei200_align_sym3, w2_200off200ma.sqw

$1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $140 \leq E \leq 150$ eV
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



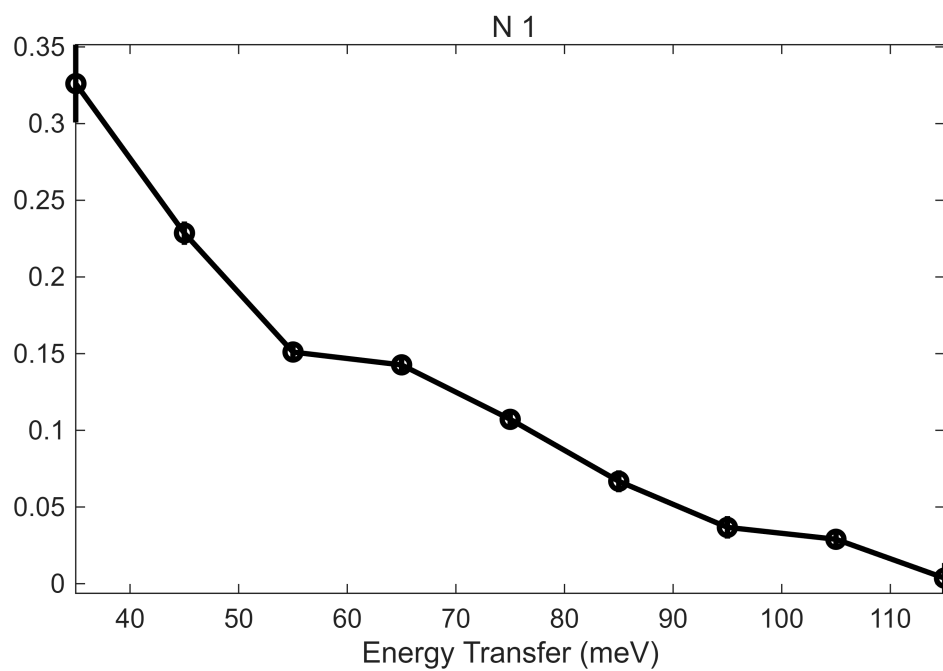




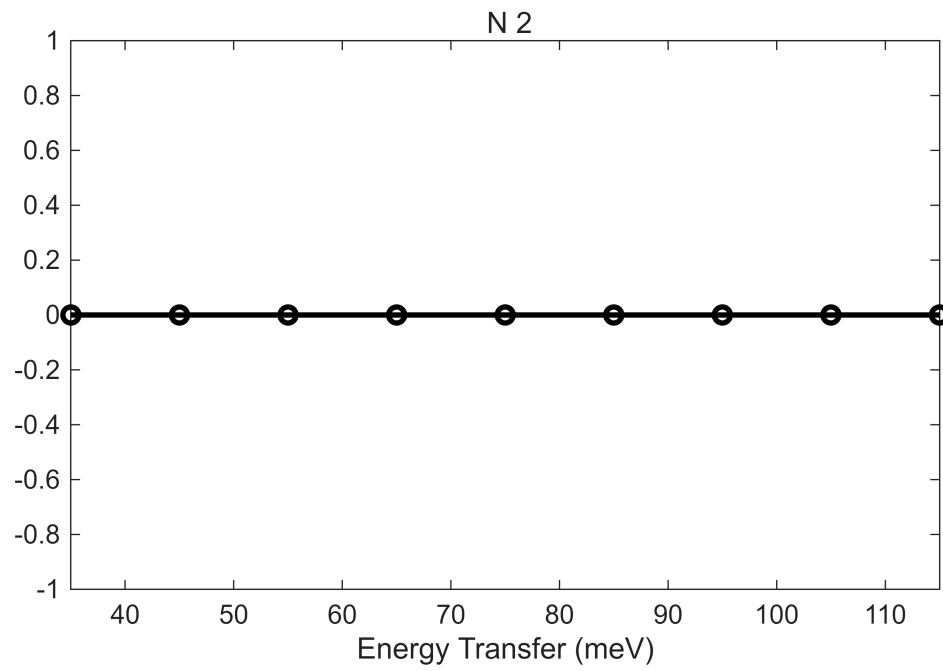
```
c1 = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```
all_fit_par = GP_FFCutsFits.Ei200_off110GPmiff_all_fit_par;
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```